# TP4 - Sécurité et Administration Distante sous Linux

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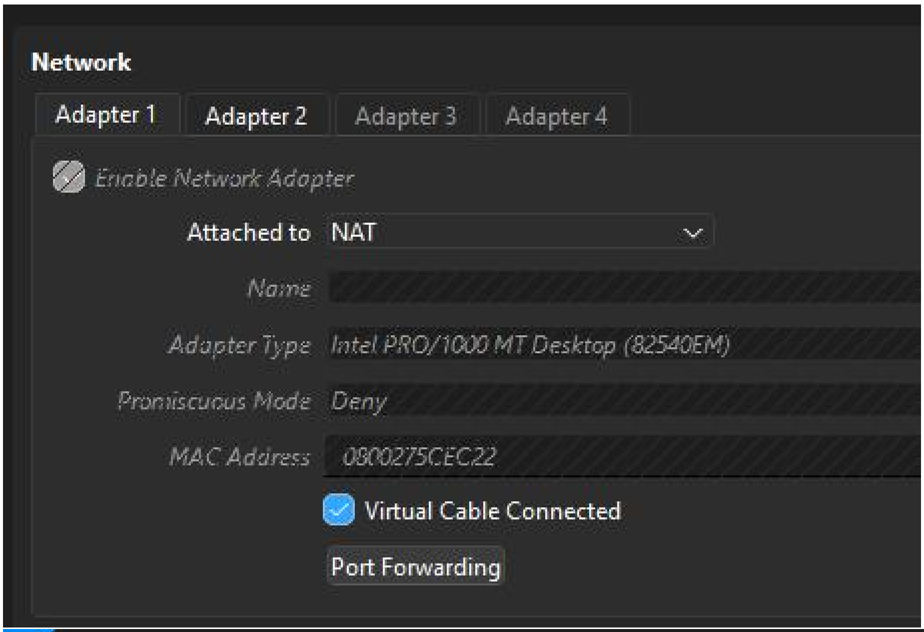
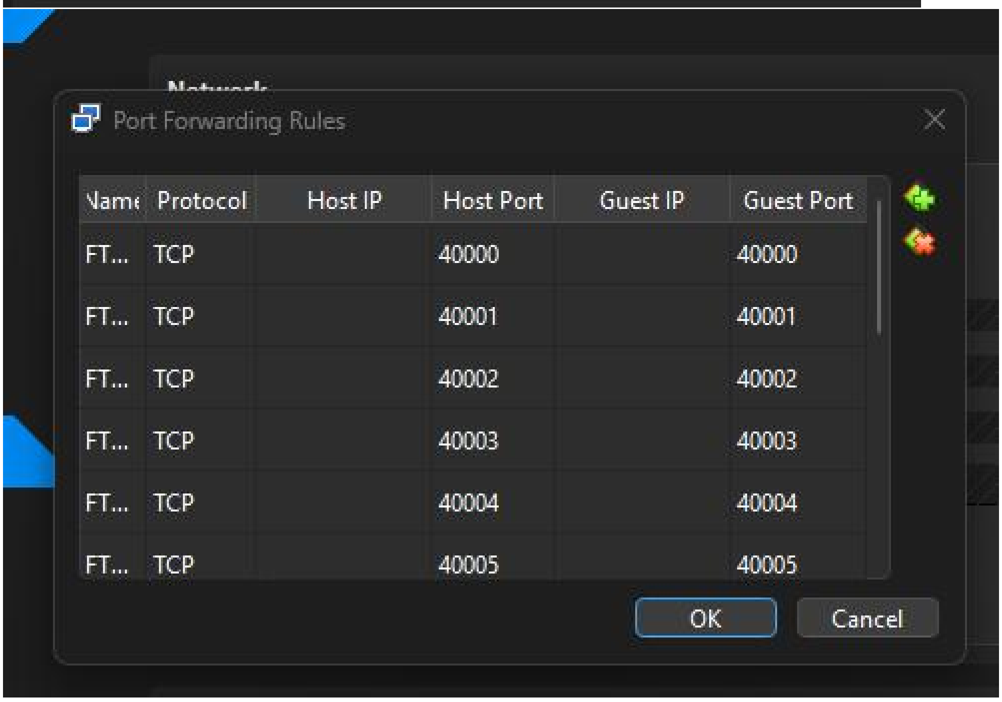
1.**Partie 1 :** Vérification de l’ infrastructure réseau existante + ajout Client Windows (NAT + Port Forwarding)



— **Partie 2 : Service FTP** :

— Contenu du fichier /etc/vsftpd.conf

```
# default to avoid remote users being able to cause excessive I/O on large
# sites. However, some broken FTP clients such as "ncftp" and "mirror" assume
# the presence of the "-R" option, so there is a strong case for enabling it.
#ls_recurse_enable=YES
#
# Customization
#
# Some of vsftpd's settings don't fit the filesystem layout by
# default.
#
# This option should be the name of a directory which is empty.  Also, the
# directory should not be writable by the ftp user. This directory is used
# as a secure chroot() jail at times vsftpd does not require filesystem
# access.
secure_chroot_dir=/var/run/vsftpd/empty
#
# This string is the name of the PAM service vsftpd will use.
pam_service_name=vsftpd
#
# This option specifies the location of the RSA certificate to use for SSL
# encrypted connections.
rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
ssl_enable=NO

#
# Uncomment this to indicate that vsftpd use a utf8 filesystem.
#utf8_filesystem=YES
#Limitation aux utilisateurs liste(optionnel)
userlist_enable=YES
userlist_deny=NO
userlist_file=/etc/vsftpd.user_list

#Mode passif (recommande derriere firewall/NAT)
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40000
```

— Statut du service vsftpd

```
server@ubuntu-server:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enab>
     Active: active (running) since Wed 2025-12-31 08:43:32 UTC; 50s ago
    Process: 8639 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, s>
   Main PID: 8641 (vsftpd)
      Tasks: 1 (limit: 2268)
     Memory: 712.0K (peak: 1.5M)
        CPU: 114ms
     CGroup: /system.slice/vsftpd.service
             └─8641 /usr/sbin/vsftpd /etc/vsftpd.conf

Dec 31 08:43:31 ubuntu-server systemd[1]: Starting vsftpd.service - vsftpd FTP ser>
Dec 31 08:43:32 ubuntu-server systemd[1]: Started vsftpd.service - vsftpd FTP serv>
```

— Tests de connexion FTP/FTPS (client Linux et client Windows)

```
ftp> ls
229 Entering Extended Passive Mode (|||40007|)
150 Here comes the directory listing.
drwxr-xr-x    2 1003     1003         4096 Dec 31 09:34 ft
p_files
226 Directory send OK.
ftp> cd ftp_files
250 Directory successfully changed.
ftp>
```

— Preuves de transfert de fichiers

```
ftp> get test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||40003|)
150 Opening BINARY mode data connection for test.txt (17 b
ytes).
100% |*************|    17       59.50 KiB/s    00:00 ETA
226 Transfer complete.
17 bytes received in 00:00 (0.66 KiB/s)
ftp>
```

```
ftp> put local_file.txt
local: local_file.txt remote: local_file.txt
229 Entering Extended Passive Mode (|||40010|)
150 Ok to send data.
100% |*************|    65        3.77 KiB/s    00:00 ETA
226 Transfer complete.
65 bytes sent in 00:00 (1.68 KiB/s)
ftp>
```

— **Partie 3 : Service SSH** :

— Configuration de /etc/ssh/sshd_config

```
  GNU nano 7.2                                                              /etc/ssh/ss

#Changer le port par default (recommande en TP)
Port 222
#Desactiver le login root par SSH
PermitRootLogin no

#Autoriser uniquement l'authentification par cles
PasswordAuthentication no
PubkeyAuthentication yes

#Desactiver l'autentification vide
PermitEmptyPasswords no

#Limiter les utilisateurs autorises
AllowUsers server ftpuser

#Protocole SSH v2 uniquement
Protocol 2

#Timeout de connexion
ClientAliveInterval 300
ClientAliveCountMax 2

#Banniere legale (optionnel)
Banner /etc/ssh/banner.txt

#Desactiver X11 Forwarding si non utilise
X11Forwarding no

#Logging
SyslogFacility AUTH
LogLevel INFO
```

— Génération de clés SSH (ssh-keygen)

```
server@ubuntu-server:~$
server@ubuntu-server:~$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
> -keyout /etc/ssl/private/vsftpd.key \
> -out /etc/ssl/certs/vsftpd.crt
..........+....+...+...........+.+...........+....+.....+++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++*...+...+.......................+......+.....
....+....+..+......+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+*.....+....+..........+....+..............+.....................+..+.+.....+...+...
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You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:
```
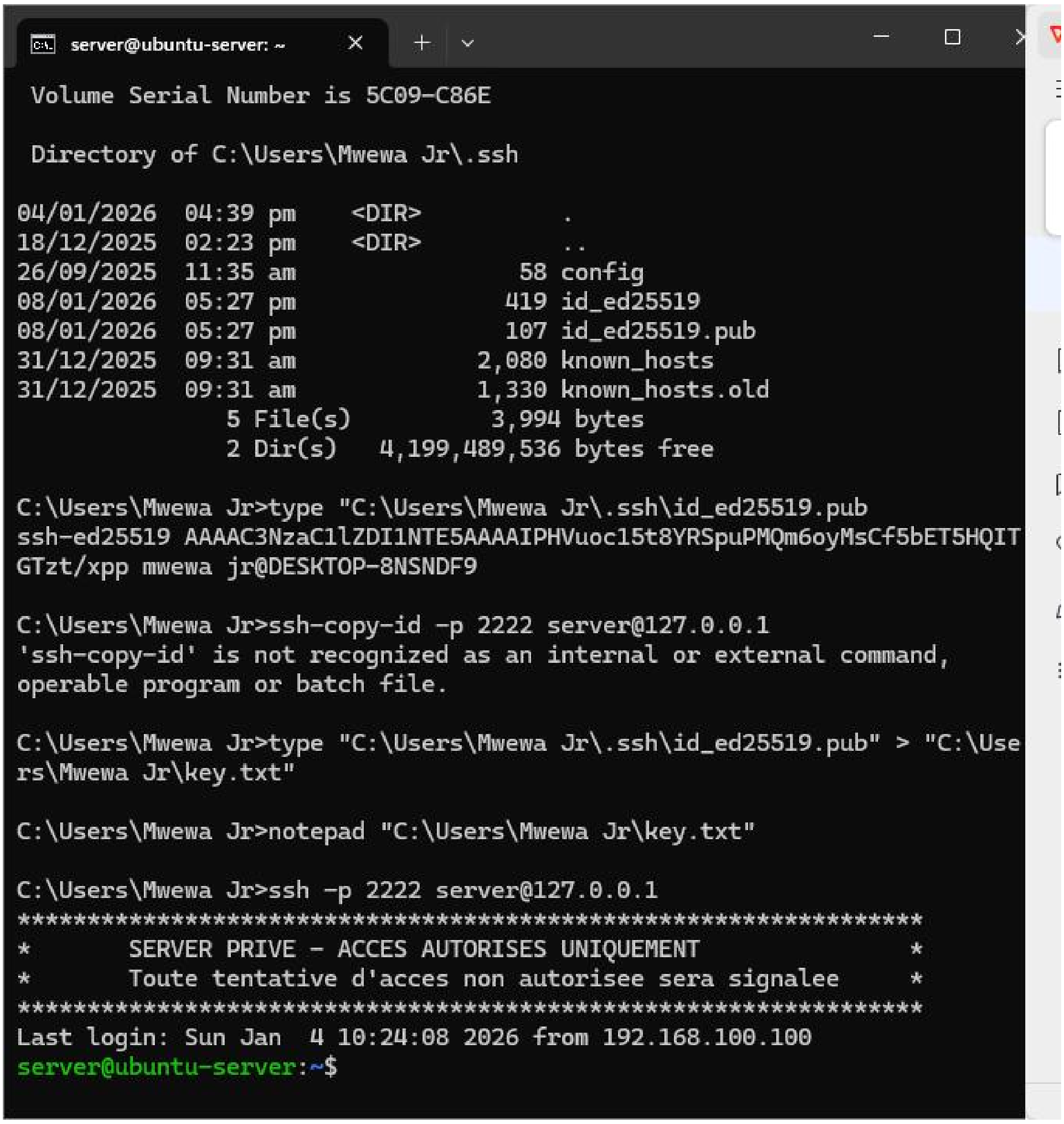
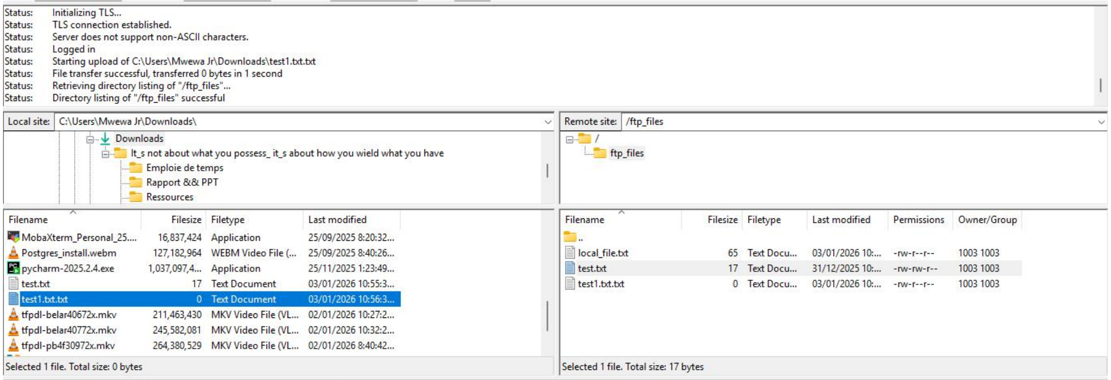

— Connexion SSH avec clés publiques (Linux et Windows)

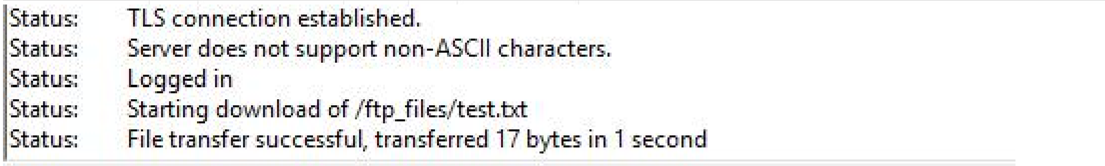
```
server@ubuntu-server:~$ ssh myserver
/home/server/.ssh/config: line 5: Bad configuration option
: identifyfile
/home/server/.ssh/config: terminating, 1 bad configuration
 options
server@ubuntu-server:~$
Warning: Permanently added '[192.168.100.10]:222' (ED25519
) to the list of known hosts.
**********************************************************
*******
*       SERVER PRIVE - ACCES AUTORISES UNIQUEMENT
      *
*       Toute tentative d'acces non autorisee sera signale
e       *
**********************************************************
*******
no such identity: /home/server/.ssh/id_rsa: No such file o
r directory
server@192.168.100.10: Permission denied (publickey,keyboa
rd-interactive).
server@ubuntu-server:~$
```

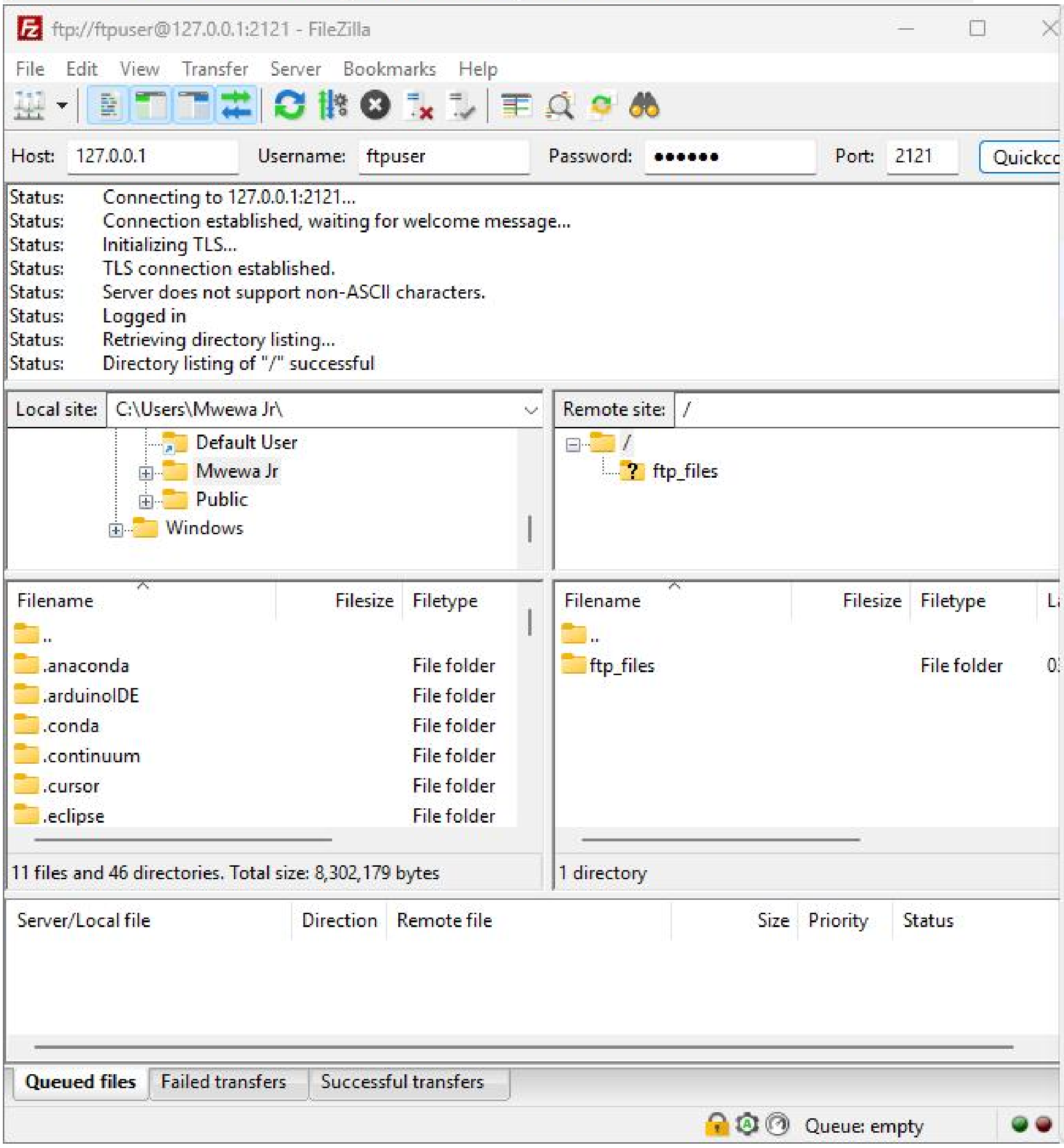


```
server@ubuntu-server: ~

 Volume Serial Number is 5C09-C86E

 Directory of C:\Users\Mwewa Jr\.ssh

04/01/2026  04:39 pm    <DIR>          .
18/12/2025  02:23 pm    <DIR>          ..
26/09/2025  11:35 am                58 config
08/01/2026  05:27 pm               419 id_ed25519
08/01/2026  05:27 pm               107 id_ed25519.pub
31/12/2025  09:31 am             2,080 known_hosts
31/12/2025  09:31 am             1,330 known_hosts.old
               5 File(s)          3,994 bytes
               2 Dir(s)   4,199,489,536 bytes free

C:\Users\Mwewa Jr>type "C:\Users\Mwewa Jr\.ssh\id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPHVuoc15t8YRSpuPMQm6oyMsCf5bET5HQIT
GTzt/xpp mwewa jr@DESKTOP-8NSNDF9

C:\Users\Mwewa Jr>ssh-copy-id -p 2222 server@127.0.0.1
'ssh-copy-id' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Mwewa Jr>type "C:\Users\Mwewa Jr\.ssh\id_ed25519.pub" > "C:\Use
rs\Mwewa Jr\key.txt"

C:\Users\Mwewa Jr>notepad "C:\Users\Mwewa Jr\key.txt"

C:\Users\Mwewa Jr>ssh -p 2222 server@127.0.0.1
*****************************************************************
*       SERVER PRIVE - ACCES AUTORISES UNIQUEMENT               *
*       Toute tentative d'acces non autorisee sera signalee     *
*****************************************************************
Last login: Sun Jan  4 10:24:08 2026 from 192.168.100.100
server@ubuntu-server:~$
```

- Tests de transfert avec scp et rsync



— Configuration du port SSH personnalisé

```
  GNU nano 7.2      /home/server/.ssh/config *
Host myserver
    HostName 192.168.100.10
    Port 222
    User server
    IdentifyFile ~/.ssh/id_rsa

Host server.lab.local
    HostName 192.168.100.10
    Port 222
    User server
```

— **Partie 4 : Pare-feu IPTABLES** :

— Liste des règles iptables (iptables -L -n -v)

```
server@ubuntu-server:~$ sudo iptables -L -n -v
[sudo] password for server:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
server@ubuntu-server:~$
server@ubuntu-server:~$
```

— Configuration des chaînes INPUT, OUTPUT, FORWARD

```
#4. aUTORISER Dhcp (server DHCP)
iptables -A INPUT -p udp --dport 67 -j ACCEPT
iptables -A OUTPUT -p udp --sport 67 -j ACCEPT

#5. Autoriser DNS (server DNS)
iptables -A INPUT -p udp --dport 53 -j ACCEPT
iptables -A INPUT -p tcp --dport 53 -j ACCEPT

#6.Autoriser http et https
iptables -A INPUT -p tcp --dport 80 -j ACCEPT
iptables -A INPUT -p tcp --dport 443 -j ACCEPT

#7.autoriser ftp (anode passif :ports 20,21,40000-40010)
iptables -A INPUT -p tcp --dport 21 -j ACCEPT
iptables -A INPUT -p tcp --dport 20 -j ACCEPT
iptables -A INPUT -p tcp --dport 40000:40010 -j ACCEPT

#8. Autoriser NFS (ports 111, 2049)
iptables -A INPUT -p tcp --dport 111 -j ACCEPT
iptables -A INPUT -p udp --dport 111 -j ACCEPT
iptables -A INPUT -p tcp --dport 2049 -j ACCEPT
iptables -A INPUT -p udp --dport 2049 -j ACCEPT


#9. Autoriser samba (ports 137-139,445)
iptables -A INPUT -p tcp --dport 139 -j ACCEPT
iptables -A INPUT -p tcp --dport 445 -j ACCEPT
iptables -A INPUT -p udp --dport 137 -j ACCEPT
iptables -A INPUT -p udp --dport 138 -j ACCEPT

#10. Autoriser icmp  (ping)
iptables -A INPUT -p icmp-type echo-request -j ACCEPT
iptables -A OUTPUT -p icmp-type echo-reply -j ACCEPT

echo "Configuration iptables apliquee avec succes"
iptables -L -n -v


server@ubuntu-server:~$
```

— Tests de filtrage (services autorisés/bloqués)

```
server@ubuntu-server: ~
.0.0.0/0            tcp dpt:445
    0     0 ACCEPT     17   --  *      *       0.0.0.0/0            0
.0.0.0/0            udp dpt:137
    0     0 ACCEPT     17   --  *      *       0.0.0.0/0            0
.0.0.0/0            udp dpt:138
    0     0 ACCEPT     1    --  *      *       0.0.0.0/0            0
.0.0.0/0            icmptype 8

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               d
estination

Chain OUTPUT (policy ACCEPT 661 packets, 56530 bytes)
 pkts bytes target     prot opt in     out     source               d
estination
    0     0 ACCEPT     0    --  *      lo      0.0.0.0/0            0
.0.0.0/0
    0     0 ACCEPT     17   --  *      *       0.0.0.0/0            0
.0.0.0/0            udp spt:67
    0     0 ACCEPT     1    --  *      *       0.0.0.0/0            0
.0.0.0/0            icmptype 0
server@ubuntu-server:~$ sudo iptables -P INPUT DROP
server@ubuntu-server:~$ sudo iptables -P FORWARD DROP
server@ubuntu-server:~$ sudo iptables -P OUTPUT ACCEPT
server@ubuntu-server:~$ sudo iptables -P INPUT ACCEPT
[sudo] password for server:
server@ubuntu-server:~$ iptables-save
iptables-save v1.8.10 (nf_tables): Could not fetch rule set generatio
n id: Permission denied (you must be root)
server@ubuntu-server:~$ sudo visudo
server@ubuntu-server:~$ iptables-save
iptables-save v1.8.10 (nf_tables): Could not fetch rule set generatio
n id: Permission denied (you must be root)
server@ubuntu-server:~$
```

— Sauvegarde des règles (iptables-save)

```
root@ubuntu-server: ~
server@ubuntu-server:~$ sudo su -
root@ubuntu-server:~# iptables-save
# Generated by iptables-save v1.8.10 (nf_tables) on Sun Jan  4 22:36:02 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [3885:320296]
-A INPUT -i lo -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A INPUT -p tcp -m tcp --dport 2222 -j ACCEPT
-A INPUT -p udp -m udp --dport 67 -j ACCEPT
-A INPUT -p udp -m udp --dport 53 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 53 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 21 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 20 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 40000:40010 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 111 -j ACCEPT
-A INPUT -p udp -m udp --dport 111 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 2049 -j ACCEPT
-A INPUT -p udp -m udp --dport 2049 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 139 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 445 -j ACCEPT
-A INPUT -p udp -m udp --dport 137 -j ACCEPT
-A INPUT -p udp -m udp --dport 138 -j ACCEPT
-A INPUT -p icmp -m icmp --icmp-type 8 -j ACCEPT
-A OUTPUT -o lo -j ACCEPT
-A OUTPUT -p udp -m udp --sport 67 -j ACCEPT
-A OUTPUT -p icmp -m icmp --icmp-type 0 -j ACCEPT
COMMIT
# Completed on Sun Jan  4 22:36:02 2026
root@ubuntu-server:~#
```

3. **Conclusion personnelle** (5-10 lignes) :

— Qu’ avez-vous appris sur la sécurité réseau ?

— Quelles difficultés avez-vous rencontrées ?

— Votre appréciation des pratiques de sécurisation

À travers ce TP, j’ai appris l’importance de la sécurité réseau dans l’administration des serveurs Linux, notamment l’usage de SSH avec authentification par clés et la configuration d’un pare-feu avec iptables. J’ai compris que chaque service exposé doit être strictement contrôlé par des règles de filtrage adaptées. Les principales difficultés rencontrées concernaient la gestion des permissions, la configuration correcte des ports (NAT, SSH, FTP) et le risque de se bloquer l’accès lors du passage en politique DROP. Ce TP m’a montré que la sécurisation demande de la rigueur, des tests fréquents et une bonne compréhension du fonctionnement réseau. Les bonnes pratiques comme le principe du moindre privilège et l’utilisation de protocoles sécurisés sont essentielles pour garantir la fiabilité d’un système.